

WEST NEEBISH ISLAND, MI, USA

WMO: 998203

Lat: 46.2847N

Lon: 84.2098W

Elev: 581

StdP: 14.39

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -14.0	(c) -8.3	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 24.2	(k) 18.2	(l) 20.9	(m) 20.8	(n) 3.4	(o) 150	(p) 0.597

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 17.9	(c) 82.2	(d) N/A	(e) 79.4	(f) N/A	(g) 76.7	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 8.1	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 23.1	(b) 19.8	(c) 17.5	(e) -23.4	(f) 87.0	(g) 8.4	(h) 2.9	(i) -29.4	(j) 89.1	(k) -34.3	(l) 90.8	(m) -39.0	(n) 92.5	(o) -45.1	(p) 94.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.4	13.9	15.4	25.2	38.1	51.0	59.5	65.4	64.5	57.6	46.4	34.9	23.3		
	DBStd	19.90	12.18	11.19	11.62	6.93	7.85	6.08	5.62	4.99	6.69	6.96	8.71	10.00		
	HDD50	4757	1118	970	768	362	83	5	0	0	10	157	456	828		
	HDD65	8779	1583	1390	1232	807	438	182	67	71	238	576	902	1293		
	CDD50	1622	0	0	1	5	114	291	476	449	236	46	4	0		
	CDD65	169	0	0	0	0	4	17	78	55	15	0	0	0		
	CDH74	1170	0	0	0	0	70	163	560	301	71	5	0	0		
	CDH80	173	0	0	0	0	9	24	102	31	7	0	0	0		
	WSAvg	7.2	7.3	7.7	8.0	8.6	7.7	6.5	6.4	6.3	6.1	7.1	7.2	7.3		
	PrecAvg	33.9	2.1	1.5	2.0	2.5	2.8	3.1	3.1	3.1	4.0	3.9	3.2	2.6		
	PrecMax	45.3	4.0	2.8	4.1	5.3	5.0	6.2	7.0	5.0	7.8	7.0	6.3	5.2		
(15) Precipitation	PrecMin	24.1	0.7	0.3	0.2	1.1	1.2	0.9	1.4	1.5	0.7	1.1	1.0	0.2		
	PrecStd	5.2	0.8	0.7	1.0	1.1	1.0	1.2	1.3	1.1	1.7	1.5	1.3	1.2		
	0.4% DB	40.9	42.8	60.3	65.5	81.4	83.5	86.3	83.9	80.4	71.6	62.5	47.1			
	0.4% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	37.1	36.1	52.1	59.3	75.0	78.2	82.8	80.4	75.7	65.8	53.0	41.9			
	2% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
	5% DB	34.6	33.7	45.9	54.3	70.9	75.2	80.0	77.7	72.4	60.8	49.7	39.0			
	5% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
	10% DB	32.2	31.7	40.9	49.9	66.2	71.7	77.0	74.9	69.1	57.0	46.4	36.6			
	10% MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
	0.4% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
	0.4% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
	2% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
	5% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
	5% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
	10% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
	10% MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
	5% DB	12.2	14.7	21.9	25.4	28.5	24.9	22.4	20.4	20.2	19.5	15.3	12.1			
	5% WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
(35) Mean Daily Temperature Range	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
	MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			
(40) Clear-Sky Solar Irradiance	taub	0.285	0.293	0.321	0.357	0.372	0.391	0.391	0.381	0.364	0.331	0.303	0.282			
	taud	2.340	2.309	2.340	2.326	2.368	2.340	2.361	2.381	2.428	2.506	2.508	2.405			
	Ebn at Noon	258	279	287	285	284	278	276	274	270	265	252	247			
	Edh at Noon	23	30	34	37	37	39	37	35	31	25	20	20			
(44) All-Sky Solar Radiation	RadAvg	357	640	1069	1430	1767	1904	1908	1614	1220	714	378	273			
	RadStd	40	71	114	154	150	137	100	104	98	81	49	25			

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page